

AI Quality Assurance Validation Suite

Document Domain: Service Operations AI Integration Testing | Status: Approved Baseline

1. System Architectural Principles & Testing Rules

These explicit behavioral constraints dictate how the automated AI service modules manage context boundaries, formatting variations, and structural fallback processes. All operational test suites verify that the model serves purely as an automated advisor while preserving strict manual overrides for the service customer.

Category Selection Constraints

The AI validation runtime engine must restrict classification mapping outputs strictly to pre-approved corporate service domains.

User Autonomy & Control

The customer retains complete authority to change, edit, reject, or ignore AI suggestions at any stage of database persistence.

Contextual Integrity Retention

The improved descriptions must expand clarity and formatting style while fully preserving the customer's core intent. The model must not introduce hypothetical symptoms, synthetic details, or external specifications missing from the client's input text.

Graceful Failure Handling

If the natural language processor enters ambiguous loops or encounters systemic exceptions, it must revert immediately to standard manual data entry forms without blocking workflow creation.

2. Functional Verification Suite: Automated Category Recommendations

Test ID	User Input Text Payload	Target Category Mapping	Expected Pipeline Output
AI-CAT-01	My kitchen sink is leaking water	Plumber	AI suggests Plumber category
AI-CAT-02	Water pipe is broken in my bathroom	Plumber	AI suggests Plumber category
AI-CAT-03	My wall socket has no power	Electrician	AI suggests Electrician category
AI-CAT-04	Need to repair broken wooden door	Carpenter	AI suggests Carpenter category
AI-CAT-05	I need to paint my house wall	Painter	AI suggests Painter category
AI-CAT-06	Cracks appeared on my cement wall	Mason	AI suggests Mason category
AI-CAT-07	My phone screen is damaged	Mobile Repair Technician	AI suggests Mobile Repair Technician category
AI-CAT-08	My refrigerator is not working	Appliance Repair Technician	AI suggests Appliance Repair Technician category
AI-CAT-09	Need to repair iron gate welding	Welder	AI suggests Welder category
AI-CAT-10	Need house deep cleaning service	Cleaner	AI suggests Cleaner category

3. Functional Verification Suite: Description Generation & Optimization

Test ID	Original Raw Text Input	Expected Enhanced Narrative Output
AI-DES-01	sink leaking	Water is leaking from the kitchen sink and requires plumbing repair.
AI-DES-02	water pipe broken	The water pipe is broken and needs repair to stop leakage.
AI-DES-03	light not working	The light is not working and requires an electrician to check the issue.
AI-DES-04	door problem	The wooden door has a problem and requires repair.
AI-DES-05	paint wall	The wall needs painting to improve its appearance.
AI-DES-06	phone broken	The mobile phone is damaged and requires repair service.
AI-DES-07	fridge not working	The refrigerator is not working and needs appliance repair.
AI-DES-08	gate damage	The metal gate is damaged and requires welding repair.
AI-DES-09	clean house	The house requires cleaning service including dust and dirt removal.
AI-DES-10	bathroom issue	The bathroom has an issue that requires inspection and repair.

4. Exception Handling & Input Validation Resilience Suite

Test ID	Malformed/Ambiguous Raw Payload	Expected Interception Response & UI Message
AI-VAL-01	[Empty String / Null String]	Show validation message: "Please enter your problem description."
AI-VAL-02	"Hi"	Intercept via fallback conversational protocol: Ask user to provide more details.
AI-VAL-03	"It is broken"	Request additional context information or assign to predefined fallback "Other" category.
AI-VAL-04	"abc123"	Show invalid standard text formatting input validation message.
AI-VAL-05	"I need help"	Trigger request loop for explicit service details required to build context.